

§2-5 Continuity

Homework : 1,2,3,4,10,16,17,18,25,37,41,43(a),(b),44,45,59,60

Definition : f is continuous at $x = a$.

$\Leftrightarrow$ The graph of f has no break at $x = a$.

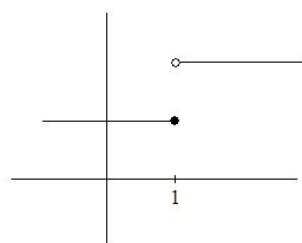
$\Leftrightarrow \lim_{x \rightarrow a} f(x) = f\left(\lim_{x \rightarrow a} x\right) = f(a)$ (極限和函數可交換).

$\Leftrightarrow$ Given $\varepsilon > 0$, $\exists$ a $\delta = \delta_\varepsilon$ s.t.

$$|f(x) - f(a)| < \varepsilon \text{ whenever } 0 < |x - a| < \delta.$$

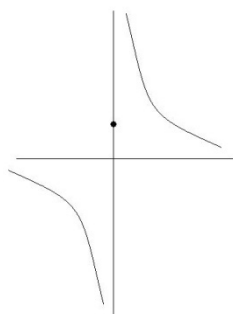
Three types of discontinuities

(i)



(jump) discontinuity at $x = 1$.

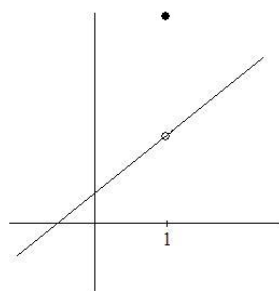
(ii)



$$f(x) = \begin{cases} \frac{1}{x}, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases}$$

(infinite) discontinuity at $x = 0$.

(iii)



$$f(x) = \begin{cases} \frac{x^2 - 1}{x - 1}, & x \neq 1 \\ 3, & x = 1 \end{cases}$$

(removable) discontinuity at $x = 1$.

Example 1 : $\lim_{x \rightarrow 1} \sqrt{x} = \sqrt{\lim_{x \rightarrow 1} x} = \sqrt{1} = 1.$

Definition : f is continuous on $I \Leftrightarrow f$ is continuous at every point in I .

- Functions that are continuous on their domains: Polynomials, Exponential, Log, Trig., Inverse Trig., Rational, Root functions.

Example 2 : $\lim_{x \rightarrow 1} \sin^{-1} \left(\frac{1 - \sqrt{x}}{1 - x} \right) = ?$

Solution : $\lim_{x \rightarrow 1} \sin^{-1} (1 + \sqrt{x}) = \sin^{-1} 2$.

Example 3 :

$$f(x) = \begin{cases} \frac{x^2 - x - 2}{x - 2}, & x \neq 2. \\ 1, & x = 2. \end{cases}$$

(i) Is f continuous at $x = 2$

(ii) If not, classify the type of discontinuity of f at $x = 2$.

Solution :

(i) $\lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x - 2} = \lim_{x \rightarrow 2} (x + 1) = 3 \Rightarrow \text{No.}$

(ii) Removable discontinuity.

Example 4 :

$$f_1(x) = [x], \quad f_2(x) = \frac{x - 7}{|x - 7|}$$

(i) Find, respectively, the set of discontinuities of f_i , $i = 1, 2$.

(ii) Classify the type of discontinuity of.

Solution :

(i) f_1 : discontinuous at all the integers.

f_2 : discontinuous at $x = 7$.

(ii) All are jump discontinuities.

Example 5 : $f(x) = \begin{cases} cx^2 + 1, & x \leq 3 \\ cx - 1, & x > 3 \end{cases}$

Find c so that f is continuous on $(-\infty, \infty)$.

Solution :

$$\begin{aligned}\lim_{x \rightarrow 3^-} f(x) &= \lim_{x \rightarrow 3^-} (cx^2 + 1) = 9c + 1 \\ \lim_{x \rightarrow 3^+} f(x) &= \lim_{x \rightarrow 3^+} (cx - 1) = 3c - 1 \\ \Rightarrow 9c + 1 &= 3c - 1 \\ \Rightarrow c &= -\frac{1}{3}\end{aligned}$$

Example 6 : $f(x) = \begin{cases} 0 & x \text{ is rational.} \\ 1 & x \text{ is irrational.} \end{cases}$

Find the set of discontinuity of f .

Solution :

因為有理數和無理數在實數線上皆稠密(dense)，因此 f 在所有實數點上皆不連續。

Example 7 : $f(x) = \begin{cases} 0 & x \text{ is rational.} \\ x & x \text{ is irrational.} \end{cases}$

Find the set of discontinuity of f .

Solution : $\lim_{x \rightarrow 0} f(x) = 0 = f(0)$. $\Rightarrow$ 在 0 點 f is continuous.

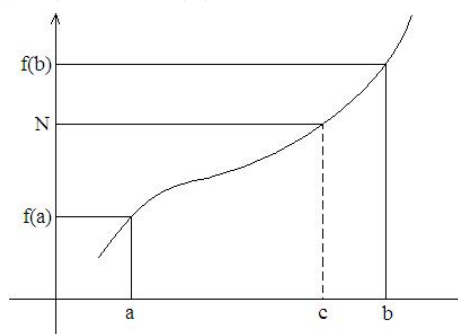
In fact, it is the only continuity.

$\Rightarrow$ Discontinuity of $f = \mathbb{R} - \{0\}$.

The Intermediate Value Theorem (中間值定理)(IVT)

- f is cont. on $[a, b]$.
- $f(a) < N < f(b)$ or $f(b) < N < f(a)$

$\Rightarrow \exists c \in (a, b)$ s.t. $f(c) = N$



Example 8 : If $f(x) = x^3 - x^2 + x$, show that $\exists a c$ s.t. $f(c) = 10$.

Solution : $f(0) = 0$, $f(3) = 21 > 10$. f is continuous on R .

$$\stackrel{(IVT)}{\implies} \exists a c \quad \text{s.t.} \quad f(c) = 10$$